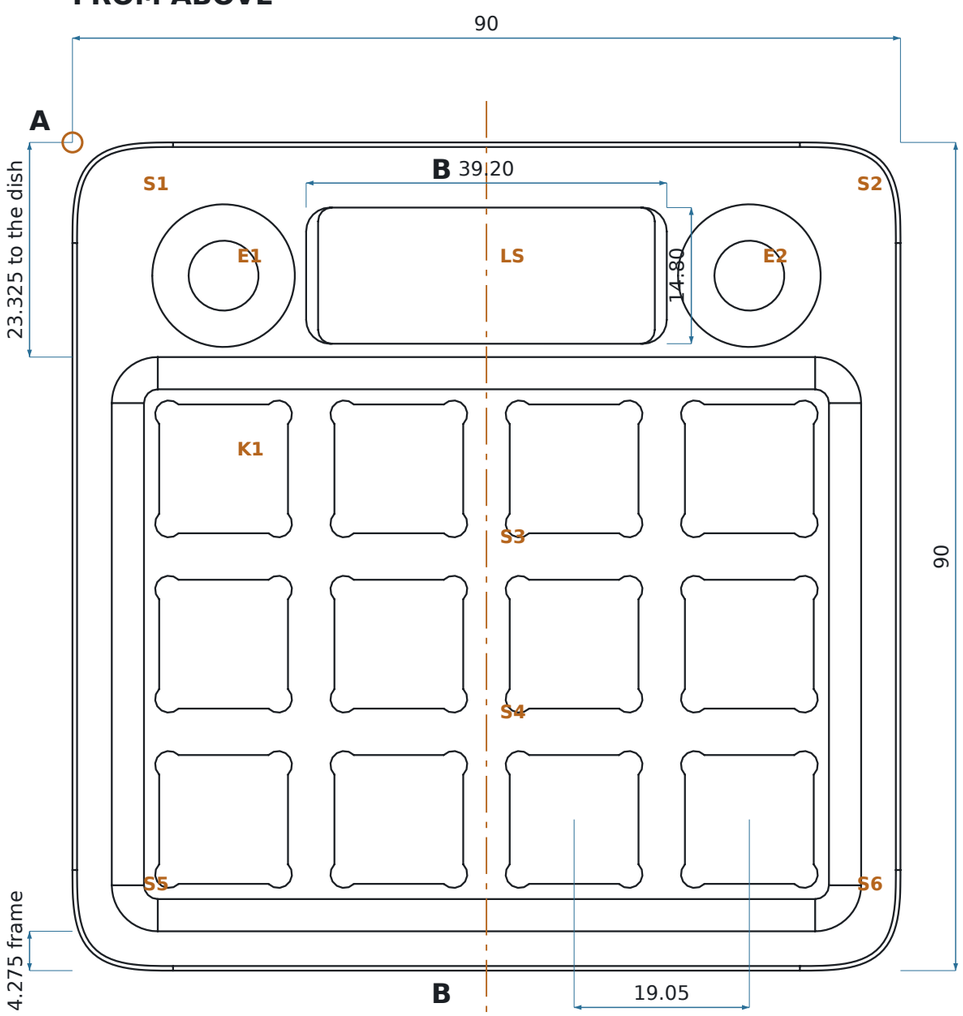
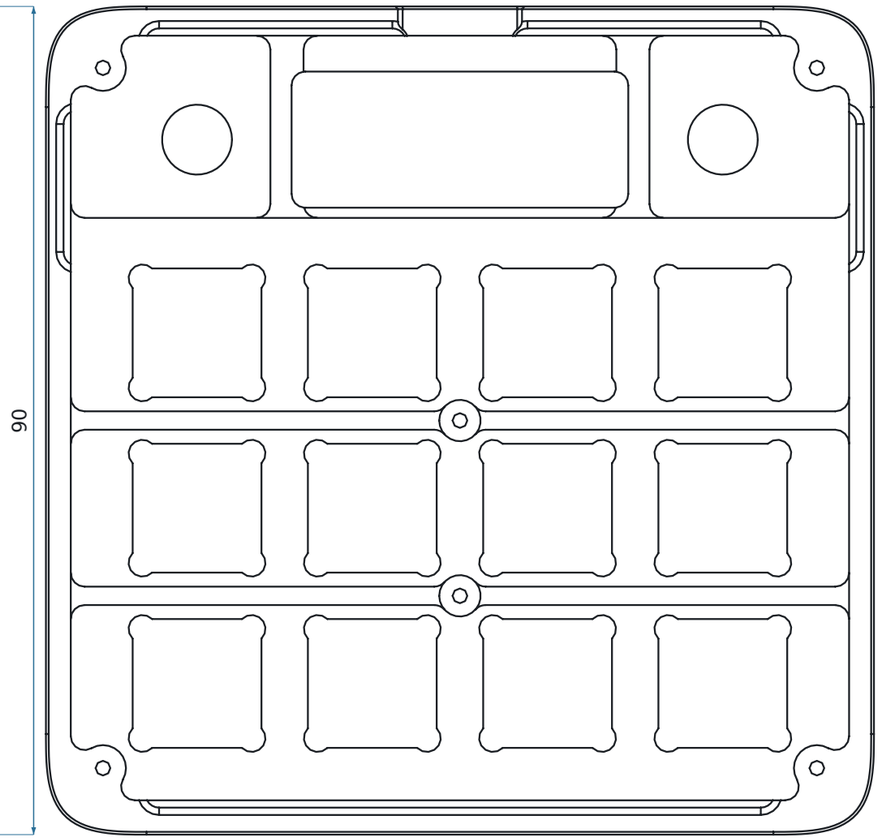
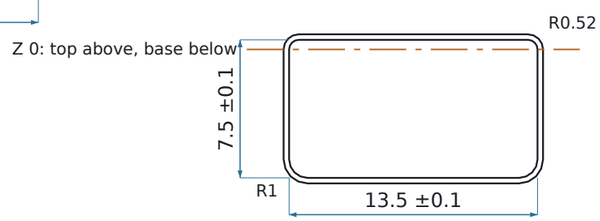
FROM ABOVE



FROM BELOW (mirrored left to right)

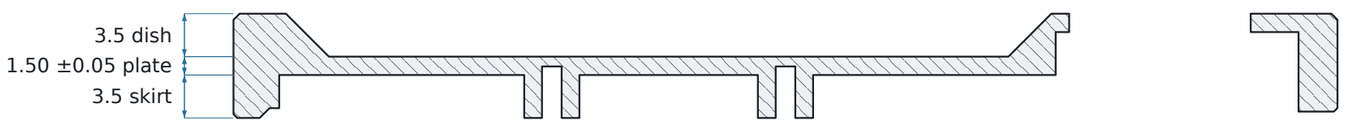




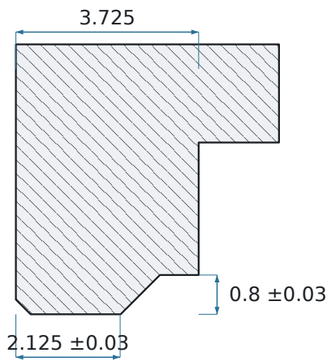
USB OPENING, 3:1

Seen from behind. Rounds along Y: R1 lower (base), R0.52 upper (top). Ball-nose or bull-nose cutter of that radius or smaller, 3D finish; the 3 mm flat end mill cannot cut them. 0.3 x 45 degree lead-in on the outer face.

SECTION B-B



GAP AND LAP JOINT, 8:1



PART	macropad case: top, CNC variant
PART NO.	MP-CASE-TOP-CNC REV A
QUANTITY	1 per macropad (order quantity on the purchase order)
MATERIAL	6061-T6 aluminium
FINISH	Bead blast, then clear anodise (natural silver)
UNITS	mm; plans seen from above and from below, as titled
TOLERANCE	ISO 2768-m unless stated; fits as listed below
STATUS	Not made or measured yet: check the first part

Holes and recesses, from datum A

tag	X	Y
M2x0.4-6H, 3.5 min full thread, from below; position ±0.05		
drill ⌀1.6 x 4.7, point to 5.18, do not exceed 4.8		
S1	6.2	6.675
S2	83.8	6.675
S5	6.2	82.775
S6	83.8	82.775
M2x0.4-6H, 3.0 min full thread, from below; position ±0.05		
drill ⌀1.6 x 4.2, flat bottom, do not exceed 4.3		
S3	45	45
S4	45	64.05
⌀7.6 through; well ⌀15.5 x 0.7 deep from above		
E1	16.425	14.475
E2	73.575	14.475
14.00 +0.05/0 square through (after anodise), first of 12 on a 19.05 grid		
K1	16.425	35.475
lens recess 39.20 +0.05/0 x 14.80 +0.05/0, R2.6		
1.25 ±0.05 deep from above; opening 36.6 x 14.8 through, in its middle		
LS	45	14.475

MATERIAL AND FINISH
6061-T6 aluminium.
Bead blast all over, 120 to 150 grit glass bead, then anodise to MIL-A-8625 Type II, Class 1 (clear, not dyed), 8 to 12 um.
Colour: natural silver, as in the renders. Cosmetic faces: the outside and the top face.
About half the coating builds up: each face grows 0.004 to 0.006 mm. The tolerances below hold after anodise.
Break unmarked edges 0.05. The 0.3 chamfers are cosmetic: measure them after blasting.
Mask the six tapped M2 holes before anodising, or chase them with an M2 6H tap after it. Rack marks: on the inner faces of the skirt, in the cavity. Not on the top face, the outside, the joint face or the lens recess.
GROUND: also mask the boss face round S1 on the joint face, ⌀5: it rests on a bare GND ring ⌀5.4 on the PCB.
TOLERANCES (after anodise)
General: ISO 2768-m.
Switch cut-outs 14.00 +0.05/0; plate 1.50 ±0.05.
S1-S6 positions from datum A: ±0.05, on both parts.
Lap grooves: depth 0.8 and land, ±0.03 (detail).
Lens recess 39.20 +0.05/0 x 14.80 +0.05/0, depth 1.25 ±0.05; USB opening ±0.1 (detail).
Flatness 0.05 on each lens ledge, 0.1 on the joint face (Z 0).
LENS OPENING
Cut it in setup 2, after the lens recess, with the same 3 mm end mill, down to Z 6.35 (2.15 deep). The recess and the opening then share their long walls without a step between setups.

Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: underside up, then top side up.
Every vertical pocket corner is R1.5 or larger; the switch cut-outs have corner reliefs of that radius. The USB opening's rounds are not vertical: see its detail.
Switch cut-outs 14.00 +0.05/0 square, measured after anodise, in a 1.50 ±0.05 mm plate.
Key dish: flat floor on the plate, walls at 45 degrees, 3.5 deep. A 90 degree chamfer mill of 8 mm or more cuts the walls.
Screw holes S1-S6: M2x0.4-6H blind, from below, for M2 x 8; depth and thread of each in the table. S3 and S4 are flat bottomed: 0.8 mm of skin is left under the dish floor.
Setup 2 (top side up): hold the part on a fixture plate through the six M2 holes, or in soft jaws cut to the outline. The skirts are too thin for vise pressure.
Outline corners: one quintic Bezier each, control points 10.94, 6.56 and 2.19 mm from the box corner on both sides. Take the curve from the STEP file.
The STEP file is the master for all geometry not dimensioned on this sheet.